

Appendix to the Climate–Nutrition Vulnerability Study

Detailed methodology, robustness tests, and supplementary tables for rice, wheat, maize and soybean

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1 Calculation methodology

1.1 Where each ingredient comes from

This table summarizes the literature basis and methodological role of each analytical component. The individual ingredients are grounded in peer-reviewed and international agency sources (FAO, WHO, IPCC), while the hazard-paired combination (F4) is our specific index design.

Ingredient	Literature backing	Status / Role in project
Heat limit T_{crit} , T_{lim} & flowering window (45–70% of season)	Deryng et al. (2014); rice: Yoshida (1981), Jagadish et al. (2007)	Published values
Water need $ET_c = K_c \times ET_0$; Hargreaves ET_0	Allen et al. (1998), Eq. 52	Published method
Water factor K_y	Steduto et al. (2012), Table 1; Brouwer and Heibloem (1986)	Published values
Signal-to-noise shift	Hawkins and Sutton (2012)	Published idea (robustness check only)
Observed yield regressions	Lobell and Field (2007); Ray et al. (2015)	Out-of-sample data check / validation
Literature sensitivity S^T, S^P	Deryng et al. (2014); Steduto et al. (2012)	Primary sensitivity scale
Importance $I = \frac{1}{5} \sum (\text{share of 5 nutrients})$	FAO FBS and USDA FoodData Central	Core index ingredient
Mean Adequacy Ratio (MAR)	WHO/FAO (2003), WHO/FAO (2004) requirements; Torheim et al. (2004)	Published adequacy metric
Multiplying E, S, I	IPCC risk framework adapted for nutrition	Methodological framing
Hazard-pairing in F4	—	Our design (Section 1.6)

Table 1: Methodological backing of ingredients.

1.2 Country panels

Country panels were constructed following fixed rules: USA, Brazil, India, and China for all four crops; one European country (Russia, top-3 in Europe for all four crops); one African producer (Nigeria for rice, Egypt for wheat, South Africa for maize and soybean); and global top-3 producers (Bangladesh for rice, Argentina for soybean). Panels represent 63% (rice), 51% (wheat), 69% (maize), and 90% (soybean) of global output.

Crop	Country	Rank	Share (%)	Class	Elev. (m)	Crop area in A/B/C/D (%)
Rice	China	1	26.4	C	260	1/1/86/12
	India	2	25.1	A	216	51/12/37/0
	Bangladesh	3	7.4	A	17	62/0/38/0
	Brazil	11	1.4	A	208	53/0/47/0
	USA	13	1.2	C	52	0/0/100/0
	Nigeria	14	1.1	A	368	89/11/0/0
	Russia	39	0.1	C	21	0/10/52/38
Wheat	China	1	17.3	C	386	0/22/51/27
	India	2	13.9	C	255	14/34/52/0
	Russia	3	11.2	D	177	0/1/2/97
	USA	4	6.1	D	647	0/31/24/44
	Egypt	17	1.2	B	42	0/100/0/0
	Brazil	19	1.0	C	567	5/0/95/0
Maize	USA	1	31.0	D	361	0/6/12/83
	China	2	23.3	D	429	0/20/28/52
	Brazil	3	9.2	A	512	53/6/41/0
	India	5	2.9	A	435	37/30/31/1
	S. Africa	8	1.3	B+M	1400	0/52/48/0
	Russia	9	1.2	D	155	0/1/12/87
Soybean	Brazil	1	36.4	A	531	59/0/41/0
	USA	2	31.5	D	278	0/0/22/77
	Argentina	3	11.4	C	159	0/6/94/0
	China	4	5.3	D	333	0/6/33/61
	India	5	3.7	A	420	70/10/20/0
	Russia	8	1.6	D	152	0/0/4/96
	S. Africa	12	0.5	C+M	1273	1/45/55/0

Table 2: Panels. Rank: global production rank; Share: % of world production; Class: primary Köppen group (+M for mountain elevation ≥ 1000 m).

1.3 Exposure

Temperature Exposure (E^T). Heat stress is evaluated over 2001–2024 using effective temperature $T_{eff} = \frac{1}{2}(T_{mean} + T_{max})$ within the flowering window (45–70% of the season length; Deryng et al. 2014):

$$E^T = \frac{L^T}{\max L^T}, \quad L^T = \text{mean}_{2001-24} \frac{\sum_m w_m \max(0, T_{eff,m} - T_{crit})}{\sum_m w_m} \quad (1)$$

where w_m is the fractional month inside the flowering period. Critical thresholds T_{crit} are: wheat 25 °C, maize 32 °C, soybean 35 °C, rice 35 °C.

Precipitation Exposure (E^P). Water stress measures the unfulfilled crop water demand on rainfed land:

$$E^P = \frac{L^P}{\max L^P}, \quad L^P = \text{mean}_{2001-24} \frac{\sum_{cells} R \max(0, 1 - P/ET_c)}{\sum_{cells} A} \quad (2)$$

where P is growing-season precipitation, $ET_c = \bar{K}_c ET_0$, R is rainfed area, and A is total harvested area. Reference evapotranspiration ET_0 is calculated with the Hargreaves equation (Allen et al., 1998) driven by extraterrestrial radiation R_a .

1.4 Sensitivity

Temperature Sensitivity (S^T). Based on the linear damage function of Deryng et al. (2014), normalized by the maximum slope across crops:

$$S^T = \frac{s^T}{\max s^T}, \quad s^T = \frac{1}{T_{lim} - T_{crit}} \quad (3)$$

For rice, $s^T = 0.07$ per °C above $T_{crit} = 35$ °C from Jagadish et al. (2007) and Yoshida (1981).

Precipitation Sensitivity (S^P). Based on FAO-66 seasonal yield response factor K_y :

$$S^P = \frac{K_y}{\max K_y} \quad (4)$$

with K_y : maize 1.25, wheat 1.10, soybean 0.85, rice 1.25. Sensitivity parameters are invariant across countries for a given crop.

Crop	Path	T_{crit}/T_{lim} (°C)	s_{heat}^{lit} (per °C)	K_y	K_c ini/mid/end	$\bar{K}_c$
Rice	C3	35 / –	0.070	1.25	1.05/1.20/0.75	1.11
Wheat	C3	25 / 35	0.100	1.10	0.40/1.15/0.33	0.88
Maize	C4	32 / 45	0.077	1.25	0.30/1.20/0.48	0.84
Soybean	C3	35 / 40	0.200	0.85	0.40/1.15/0.50	0.89

Table 3: Literature parameters used in Exposure and Sensitivity.

1.5 Importance and Nutrient Adequacy

Dietary Importance (I) is the unweighted average national supply share of five essential nutrients:

$$I_{kc} = \frac{1}{5} \sum_{n \in \{\text{kcal, protein, fat, Zn, Fe}\}} \frac{\text{supply}_n \text{ from crop } k}{\text{total supply}_n}, \quad (\text{FBS 2021–23}) \quad (5)$$

Zinc and iron supply is derived by multiplying item-level food quantities by USDA FoodData Central composition values. Adequacy is calculated as $\text{NAR}_n = \min(1, \text{supply}_n / \text{requirement}_n)$ and $\text{MAR} = \frac{1}{5} \sum_n \text{NAR}_n$.

1.6 Candidate Formulas and Choice of F4

Four candidate aggregation forms were evaluated:

$$\text{F1 Multiplicative: } V = E \cdot S \cdot I, \quad E = \frac{1}{2}(E^T + E^P), \quad S = \frac{1}{2}(S^T + S^P) \quad (6)$$

$$\text{F2 Geometric: } V = (E \cdot S \cdot I)^{1/3} \quad (7)$$

$$\text{F3 Additive: } V = (E + S + I)/3 \quad (8)$$

$$\text{F4 Hazard-Paired: } V = \frac{1}{2}(E^T S^T + E^P S^P) I \quad (9)$$

Formula F4 is chosen because:

- It prevents cross-hazard mismatch (e.g. heat exposure multiplying water sensitivity).

- Crops with near-zero direct food use ($I \approx 0$) correctly receive near-zero nutritional vulnerability scores.
- Its climate component reflects the expected relative yield loss under observed hazards.

Form	Country weights	Rice	Wheat	Maize	Soybean
F1 $E \cdot S \cdot I$	equal	0.001 (4)	0.064 (1)	0.021 (2)	0.010 (3)
	production	0.001 (4)	0.081 (1)	0.006 (3)	0.012 (2)
F2 $(ESI)^{1/3}$	equal	0.077 (4)	0.366 (1)	0.201 (2)	0.192 (3)
	production	0.081 (4)	0.407 (1)	0.153 (3)	0.207 (2)
F3 $(E + S + I)/3$	equal	0.229 (4)	0.410 (1)	0.318 (2)	0.297 (3)
	production	0.283 (3)	0.429 (1)	0.242 (4)	0.295 (2)
F4 paired $\times I$	equal	0.002 (4)	0.065 (1)	0.018 (2)	0.008 (3)
	production	0.001 (4)	0.084 (1)	0.005 (3)	0.009 (2)

Table 4: Crop vulnerability index across formulas and weighting methods.

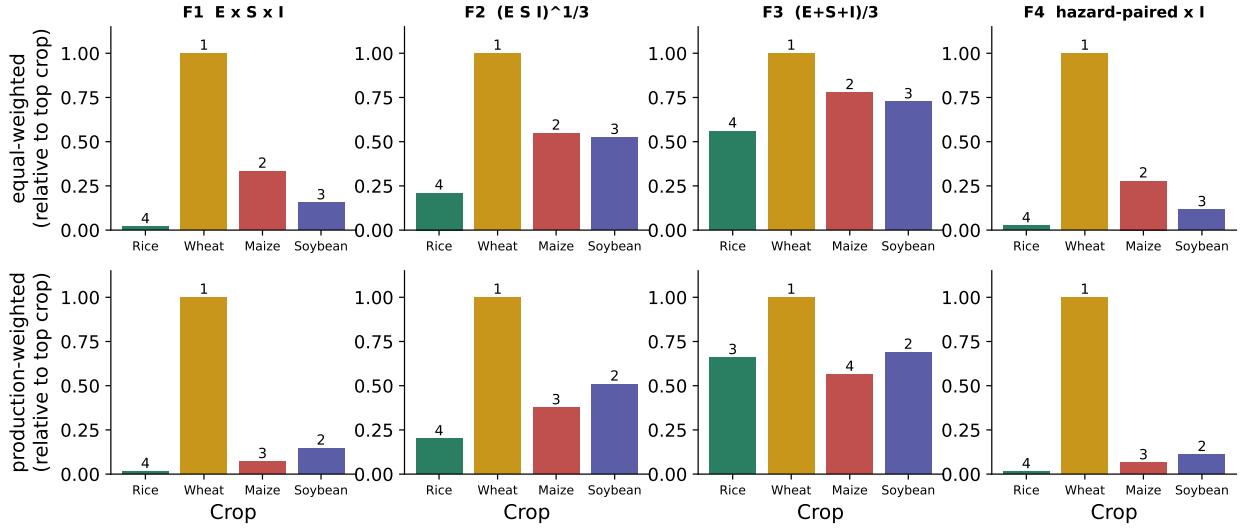


Figure 1: Crop vulnerability across candidate formulas relative to the maximum crop score.

2 Supplementary tables

Crop	Country	Heat		Water		Exposure		Sensitivity		I (%)	V_{F4}
		L (°C)	Δ (SD)	L (%)	Δ (SD)	E_h	E_w	S_h	S_w		
Rice	China (C)	0.00	0.77	0.0	0.00	0.00	0.00	0.17	0.50	13.3	0.000
	India (A)	0.00	1.05	1.3	0.00	0.00	0.02	0.66	0.50	15.6	0.002
	Bangladesh (A)	0.00	0.87	0.1	0.12	0.00	0.00	0.17	0.50	42.1	0.000
	Brazil (A)	0.00	1.08	7.6	0.06	0.00	0.14	0.17	0.50	4.7	0.004
	USA (C)	0.00	0.46	0.0	0.00	0.00	0.00	0.33	0.50	1.2	0.000
	Nigeria (A)	0.00	0.64	7.6	0.00	0.00	0.14	0.17	0.50	5.6	0.005
	Russia (C)	0.00	1.58	0.2	0.00	0.00	0.00	0.17	0.50	0.9	0.000
Wheat	China (C)	0.01	1.34	15.0	0.01	0.02	0.27	0.37	0.44	16.0	0.024
	India (C)	0.58	0.90	17.8	0.02	1.00	0.33	0.43	0.44	22.2	0.151
	Russia (D)	0.07	0.54	31.4	0.05	0.13	0.57	0.37	0.44	31.0	0.111
	USA (D)	0.08	0.24	36.9	0.01	0.13	0.68	0.37	0.44	18.2	0.075
	Egypt (B)	0.08	1.46	0.1	0.00	0.14	0.00	0.37	0.44	34.4	0.021
	Brazil (C)	0.01	0.96	7.3	0.25	0.02	0.13	0.37	0.44	13.3	0.010
Maize	USA (D)	0.01	0.38	16.3	0.00	0.02	0.30	0.46	0.50	2.0	0.004
	China (D)	0.01	0.77	8.5	0.00	0.01	0.15	0.40	0.66	1.3	0.002
	Brazil (A)	0.00	0.99	13.6	0.21	0.01	0.25	0.69	0.50	6.7	0.010
	India (A)	0.04	0.85	3.7	0.00	0.06	0.07	0.69	0.50	1.6	0.001
	S. Africa (B+M)	0.00	0.86	36.5	0.26	0.00	0.67	0.69	0.50	22.5	0.089
	Russia (D)	0.00	1.20	54.6	0.48	0.00	1.00	0.69	1.00	0.2	0.002
Soybean	Brazil (A)	0.00	1.08	3.8	0.13	0.00	0.07	0.76	0.42	7.3	0.002
	USA (D)	0.00	0.35	22.3	0.00	0.00	0.41	0.67	0.42	9.7	0.020
	Argentina (C)	0.00	0.55	26.1	0.19	0.00	0.48	0.76	0.61	1.5	0.005
	China (D)	0.00	0.71	13.8	0.00	0.00	0.25	0.76	0.56	5.8	0.010
	India (A)	0.00	0.59	0.9	0.00	0.00	0.02	0.76	0.42	2.4	0.000
	Russia (D)	0.00	0.63	32.5	0.09	0.00	0.60	0.76	0.42	1.1	0.003
	S. Africa (C+M)	0.00	0.83	35.4	0.33	0.00	0.65	0.76	0.42	3.8	0.012

Table 5: Index components for each crop–country pair (L : level 2001–24; Δ : shift in SDs; E^T, E^P scaled 0–1; S^T, S^P scaled 0–1; I : nutritional share; V_{F4} : vulnerability index).

Country	Rice	Wheat	Maize	Soybean
USA	< 0.0001	0.0747	0.0037	0.0197
Brazil	0.0039	0.0103	0.0103	0.0025
India	0.0023	0.1511	0.0015	0.0002
China	0.0001	0.0240	0.0016	0.0097
Russia	< 0.0001	0.1105	0.0021	0.0032
Argentina	–	–	–	0.0052
Bangladesh	0.0002	–	–	–
Egypt	–	0.0209	–	–
Nigeria	0.0046	–	–	–
South Africa	–	–	0.0895	0.0122

Table 6: Country-by-crop F4 vulnerability index values (V_{F4}).

Crop	Country	Heat: % yield per +1°C T_{eff}				Water: % yield per +0.1 MR				R^2
		lin	quad	FD	used	lin	quad	FD	used	
Rice	China	-1.4	-0.2	-0.9	+0.0 (p)	-0.5	+0.3	+0.6	+0.0 (p)	0.04
	India	-10.9	-10.8	-12.4	-11.4 (c)	+1.7	+1.7	+2.0	+0.0 (p)	0.26
	Bangladesh	-5.1	-4.7	-4.1	+0.0 (p)	-0.0	-0.0	-0.0	+0.0 (p)	0.02
	Brazil	-15.7	-2.7	-5.2	+0.0 (p)	+2.2	+2.2	+2.1	+0.0 (p)	0.01
	USA	-3.9	-3.8	-2.9	-3.5 (c)	+0.0	+0.0	+0.0	+0.0 (p)	0.26
	Nigeria	+3.4	+5.4	+6.7	+0.0 (p)	-4.3	-0.1	+0.9	+0.0 (p)	0.04
	Russia	+6.0	-5.1	-3.2	+0.0 (p)	-4.3	-18.0	-13.8	+0.0 (p)	0.17
Wheat	China	-1.8	-2.0	+2.0	-2.7 (p)	+0.3	+0.2	-1.7	+0.0 (p)	0.02
	India	-4.6	-4.0	-4.1	-4.2 (c)	-2.8	-1.4	+0.9	+0.0 (p)	0.13
	Russia	+0.2	-1.2	+0.5	-2.7 (p)	+5.5	+1.9	+6.8	+0.0 (p)	0.19
	USA	-0.1	-0.1	-0.3	-2.7 (p)	+1.8	+1.8	+0.7	+0.0 (p)	0.13
	Egypt	+0.9	+0.2	-1.0	-2.7 (p)	+2.1	+2.9	+0.2	+0.0 (p)	0.04
	Brazil	-10.7	-11.3	-14.9	-2.7 (p)	-3.1	-3.3	-5.0	-3.8 (c)	0.22
Maize	USA	-5.9	-5.9	-7.3	-6.4 (c)	+0.7	+0.7	+0.2	+0.0 (p)	0.34
	China	-5.5	-5.0	-4.3	-4.9 (c)	+3.6	+3.8	+3.5	+3.6 (c)	0.22
	Brazil	-2.6	-3.2	-2.5	-11.7 (p)	+2.1	+2.1	+2.0	+0.0 (p)	0.05
	India	-11.2	-10.4	-10.8	-11.7 (p)	+0.3	+0.3	+0.1	+0.0 (p)	0.09
	S. Africa	-25.2	-27.2	-31.1	-27.8 (c)	-3.0	-4.5	-1.9	+0.0 (p)	0.27
	Russia	+6.7	+7.6	+11.3	-11.7 (p)	+29.1	+33.5	+36.5	+33.0 (c)	0.50
Soybean	Brazil	-4.0	-3.4	-3.1	-6.2 (p)	+2.8	+3.0	+3.2	+1.8 (p)	0.05
	USA	-3.9	-3.9	-4.0	-3.9 (c)	+1.1	+1.2	+1.5	+1.8 (p)	0.26
	Argentina	+0.4	-2.3	-4.9	-6.2 (p)	+7.0	+6.7	+5.3	+6.3 (c)	0.54
	China	+4.0	+1.4	+4.2	-6.2 (p)	+5.0	+5.1	+5.2	+5.1 (c)	0.05
	India	-16.1	-15.3	-10.6	-6.2 (p)	+1.0	+0.9	+0.5	+1.8 (p)	0.28
	Russia	-17.6	-10.1	-8.2	-6.2 (p)	+14.4	+18.1	+16.9	+1.8 (p)	0.10
	S. Africa	-17.5	-11.4	-15.1	-6.2 (p)	-5.7	+1.1	-2.6	+1.8 (p)	0.17

Table 7: Out-of-sample empirical yield regressions over 1971–2000 (lin/quad/FD detrending; used: coefficient entering the data check; R^2 of linear model).

Crop	Country	FBS 1961–2013	FBS 2010–2023	MAR pts per 1% yield
Rice	China	+0.90*	-0.31	-0.000
	India	+0.37*	+0.54	+0.025
	Bangladesh	+0.37*	-0.36	-0.219
	Brazil	-0.07*	+0.51	-0.027
	USA	-0.07	-0.43	-0.000
	Nigeria	+0.28	-0.22	-0.038
	Russia	+0.21	-0.83	-0.000
Wheat	China	+0.62	-0.79*	-0.000
	India	+0.54*	-0.06	+0.017
	Russia	-0.04	+0.09	-0.000
	USA	-0.19	-0.04	-0.000
	Egypt	-0.35*	-0.02	-0.000
	Brazil	-0.05	+0.19	+0.007
Maize	USA	-0.13	-0.01	-0.000
	China	+0.29	-0.38	-0.000
	Brazil	+0.19*	-0.03	-0.010
	India	+0.40*	-1.97	+0.049
	S. Africa	-0.00	-0.22	-0.003
	Russia	+0.31	-0.82	-0.000
Soybean	Brazil	-0.65	+0.00	+0.000
	USA	-0.40*	-0.23*	-0.000
	Argentina	+1.59*	+0.03	+0.004
	China	+0.10	+0.37	+0.000
	India	-1.84	-0.09	+0.009
	Russia	+1.56*	+3.04*	+0.000
	S. Africa	+1.83*	+0.08	-0.003

Table 8: Yield pass-through to food supply and MAR (FBS 1961–2013 and 2010–2023).

Crop	Country	SSR	Climate-driven ΔY (%)		Standard -10% yield shock (S3)				$\Delta \text{MAR pts}$ (no trade / trade)
			S1 shift	S2 1-in-10	$\Delta \text{kcal}\%$	$\Delta \text{Zn}\%$	$\Delta \text{Fe}\%$		
Rice	China	0.95	+0.0	+0.0	-2.36	-1.51	-0.82		+0.00 / +0.00
	India	1.19	-4.8	-3.3	-2.47	-2.09	-1.15		-0.20 / -0.00
	Bangladesh	1.02	+0.0	+0.0	-6.29	-5.45	-3.62		-0.50 / -0.51
	Brazil	1.04	+0.0	+0.0	-0.78	-0.59	-0.40		-0.08 / +0.00
	USA	1.43	-1.4	-2.7	-0.21	-0.15	-0.11		+0.00 / +0.00
	Nigeria	1.00	+0.0	+0.0	-0.83	-0.76	-0.37		-0.07 / -0.07
	Russia	0.79	+0.0	+0.0	-0.12	-0.09	-0.06		+0.00 / +0.00
Wheat	China	0.90	-3.7	-4.0	-1.62	-1.92	-1.72		+0.00 / +0.00
	India	1.01	-3.2	-4.1	-2.11	-3.26	-2.95		-0.52 / -0.28
	Russia	1.78	-3.1	-4.8	-2.91	-4.56	-4.90		+0.00 / +0.00
	USA	1.25	-1.1	-4.8	-1.54	-2.72	-3.09		+0.00 / +0.00
	Egypt	0.56	-3.6	-3.3	-1.90	-2.70	-2.57		+0.00 / +0.00
	Brazil	0.69	+1.1	-16.8	-0.75	-1.41	-1.57		-0.31 / +0.00
Maize	USA	1.20	-2.8	-4.6	-0.21	-0.29	-0.33		+0.00 / +0.00
	China	0.92	-2.8	-4.4	-0.15	-0.16	-0.14		+0.00 / +0.00
	Brazil	1.55	-7.0	-6.1	-0.72	-0.92	-1.05		-0.20 / +0.00
	India	1.15	-4.6	-2.7	-0.18	-0.23	-0.21		-0.04 / -0.00
	S. Africa	1.36	-26.8	-23.8	-2.17	-2.95	-3.23		-0.53 / -0.00
	Russia	1.28	-38.1	-31.4	-0.02	-0.03	-0.03		+0.00 / +0.00
Soybean	Brazil	2.52	-4.3	-4.3	-0.98	-0.00	-0.01		-0.00 / +0.00
	USA	1.65	-0.8	-3.9	-1.43	-0.00	-0.01		+0.00 / +0.00
	Argentina	0.97	-4.6	-16.9	-0.20	-0.00	-0.00		+0.00 / +0.00
	China	0.27	-2.6	-7.6	-0.10	-0.08	-0.18		+0.00 / +0.00
	India	0.82	+0.8	-8.9	-0.17	-0.01	-0.03		-0.01 / -0.01
	Russia	1.02	-4.4	-4.9	-0.13	-0.00	-0.01		+0.00 / +0.00
	S. Africa	1.13	-7.1	-6.6	-0.43	-0.03	-0.09		-0.01 / -0.00

Table 9: Nutritional impact under yield shock scenarios (S1 observed shift, S2 1-in-10 shock, S3 -10% standard shock).

3 Supplementary figures

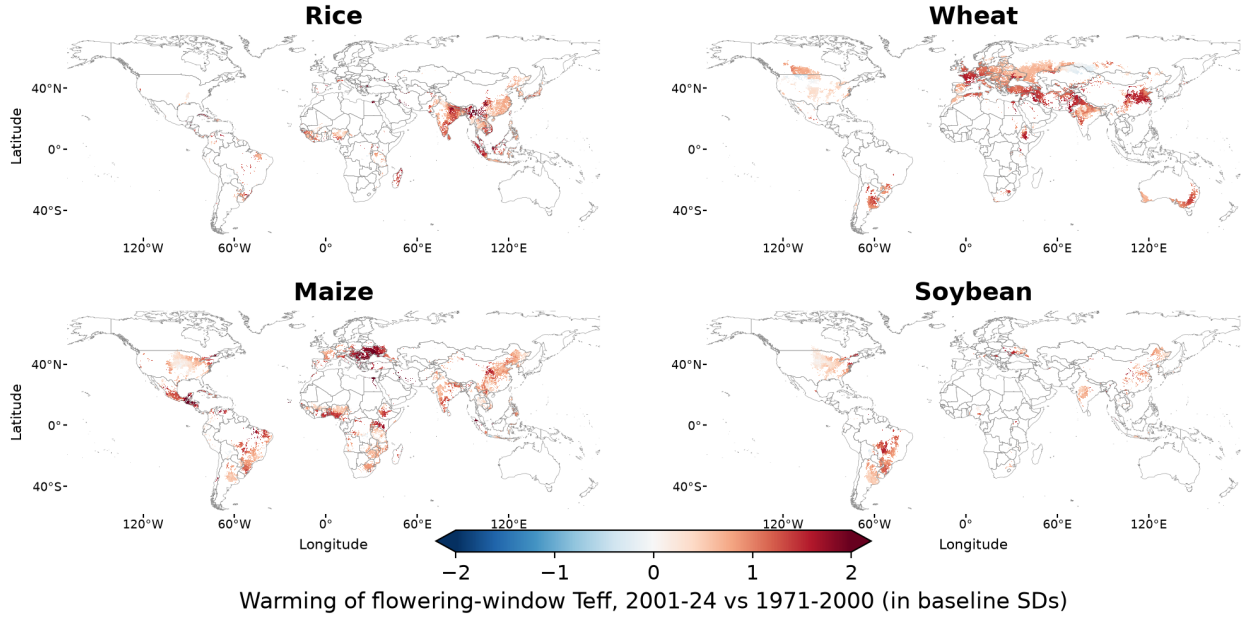


Figure 2: Warming of the flowering season (2001–2024 vs 1971–2000 baseline SDs).

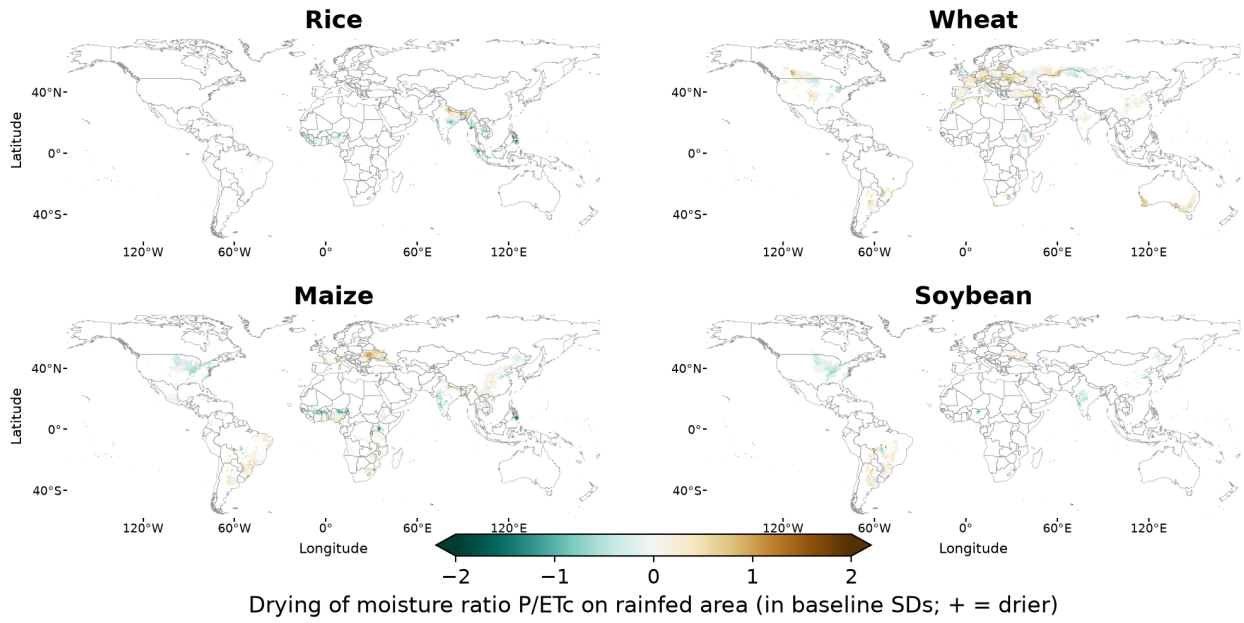


Figure 3: Drying of the rainfed growing season (shift in moisture ratio P/ET_c in baseline SDs).

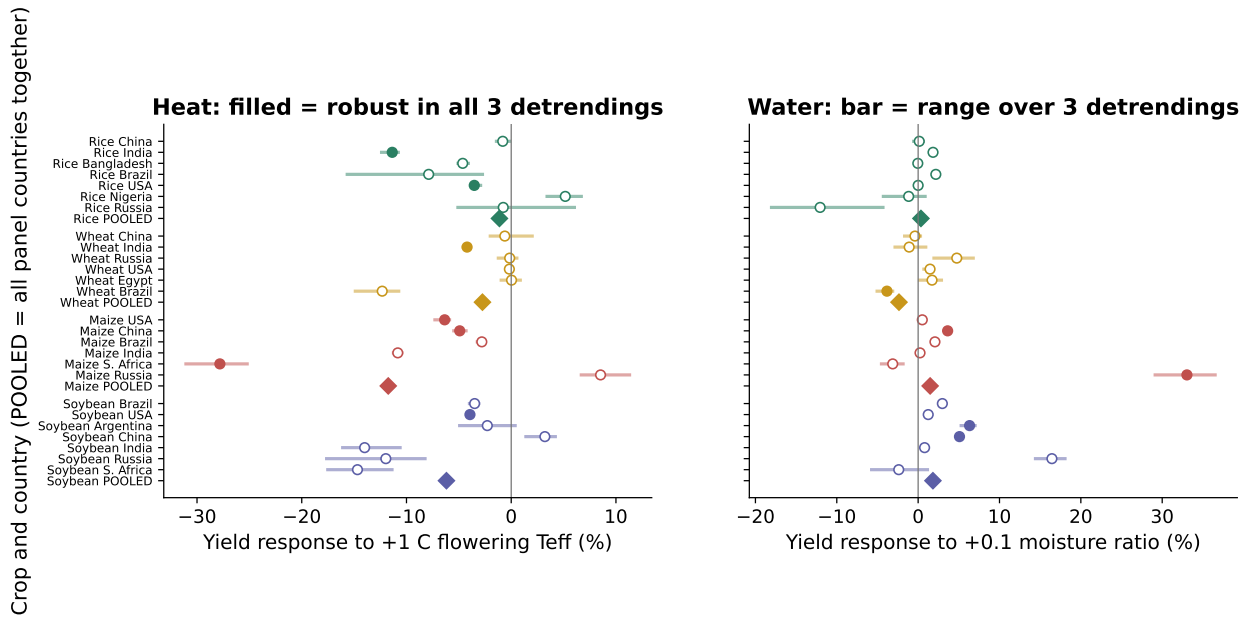


Figure 4: Observed yield sensitivity coefficients by crop and country.

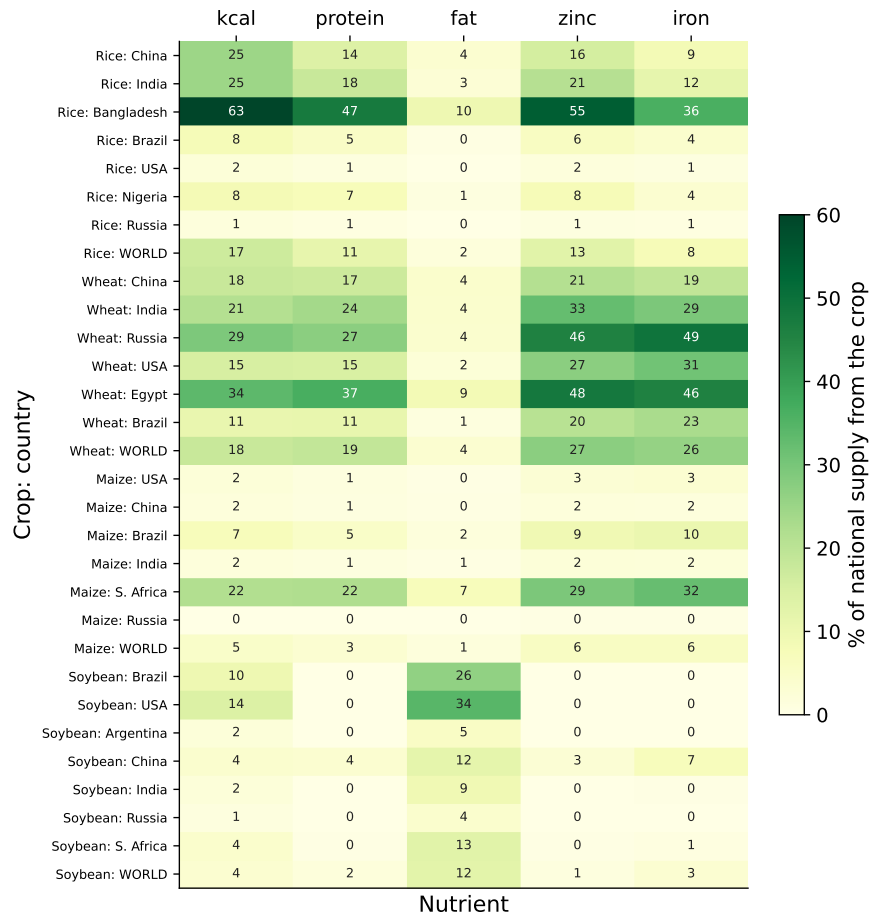


Figure 5: Nutritional supply share heatmap across all 5 nutrients by crop and country.

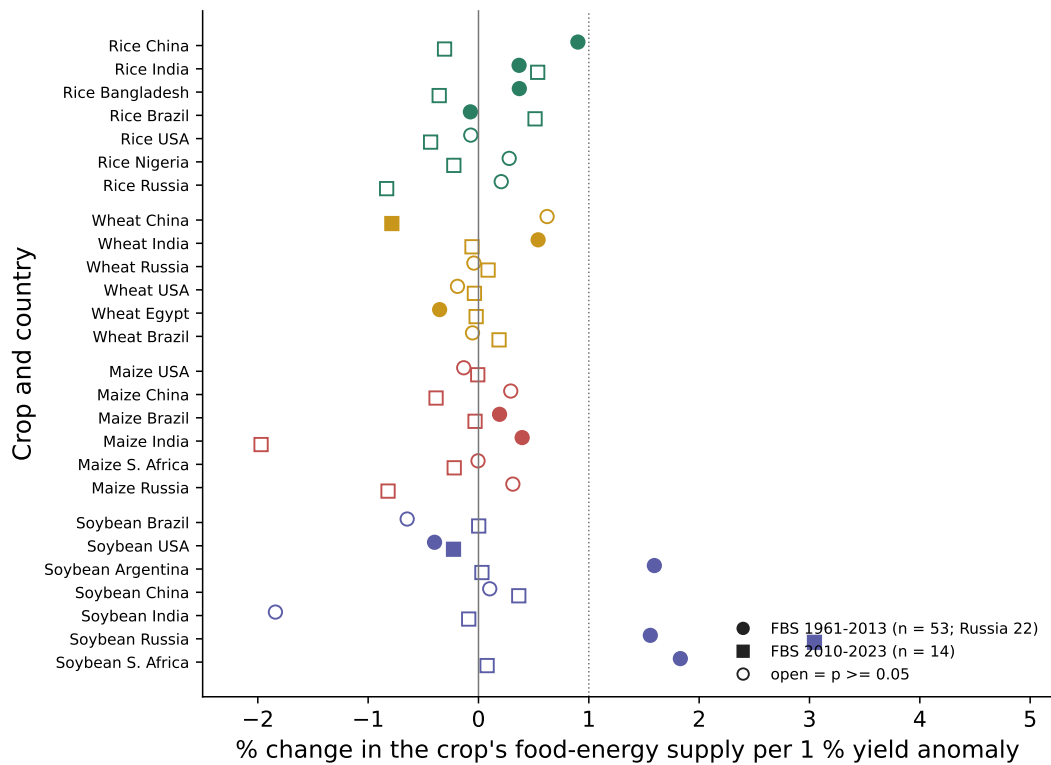


Figure 6: Yield-to-supply pass-through regression coefficients across countries.

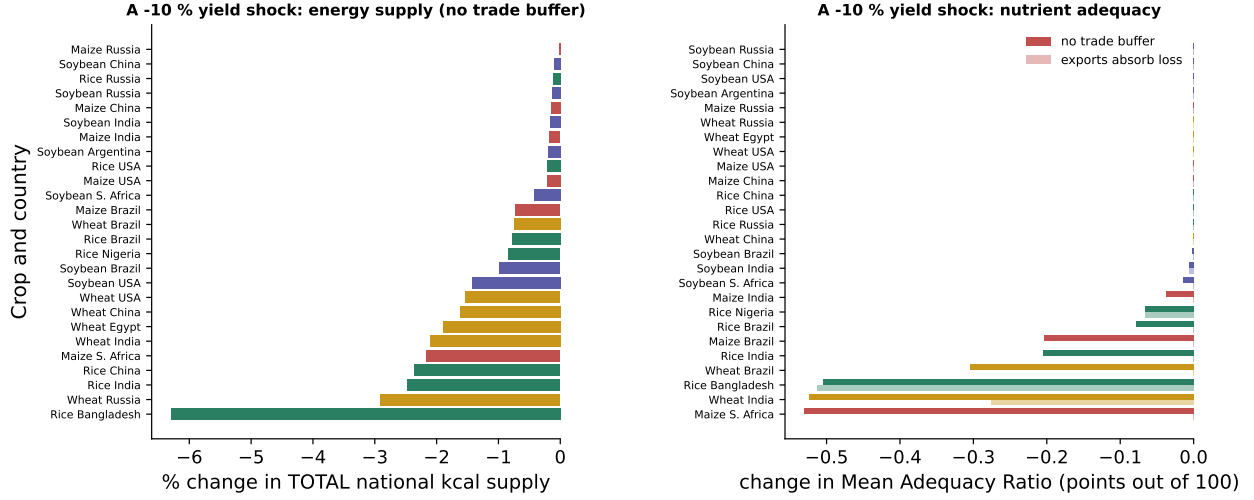


Figure 7: Impact of a -10% yield shock on calories and Mean Adequacy Ratio (MAR).

4 Robustness and validation

4.1 Robustness

A series of 10 one-at-a-time modifications was evaluated (min-max normalization, level vs shift exposure, literature vs empirical sensitivity modes, rice K_y proxies, and calorie-only vs micronutrient-only importance). Across all 80 tests, **wheat is ranked #1 in 80 cases**. Monte Carlo simulation (2,000 Dirichlet draws) places wheat first in 97.5% of iterations under F4.

Test (one choice changed)	F4 hazard-paired order	F1 multiplicative order
main (x/max, exposure = stress level, lit+emp S)	wheat > maize > soybean > rice	wheat > maize > soybean > rice
min-max normalisation	wheat > maize > soybean > rice	wheat > maize > soybean > rice
exposure = level + change (average)	wheat > maize > rice > soybean	wheat > maize > rice > soybean
exposure = change only	wheat > maize > rice > soybean	wheat > maize > rice > soybean
sensitivity = literature only	wheat > maize > soybean > rice	wheat > maize > soybean > rice
sensitivity = empirical only	wheat > soybean > maize > rice	wheat > maize > soybean > rice
rice $K_y = 1.0$	wheat > maize > soybean > rice	wheat > maize > soybean > rice
rice $K_y = 1.5$	wheat > maize > soybean > rice	wheat > maize > soybean > rice
importance = kcal share only	wheat > maize > soybean > rice	wheat > maize > soybean > rice
importance = Zn+Fe+protein only	wheat > maize > rice > soybean	wheat > maize > soybean > rice

Table 10: Crop ranking under each one-at-a-time robustness test.

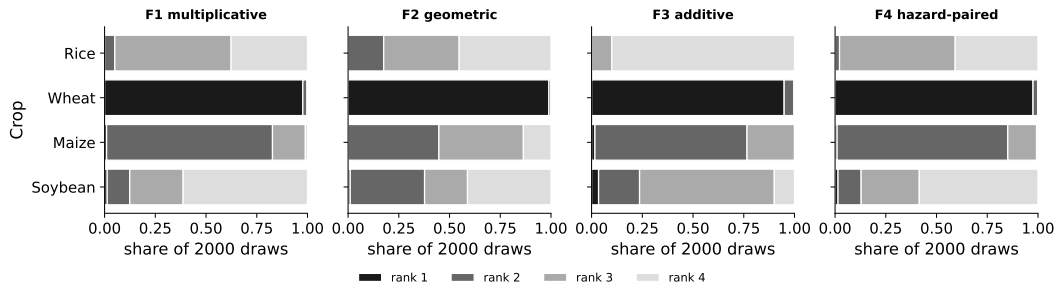


Figure 8: Rank distribution from 2,000 Monte Carlo draws of component sub-weights.

4.2 Validation Benchmarks

- **Zhao et al. (2017):** Comparison against published multi-model warming yield loss shows rank agreement $\rho = +0.6$.
- **Myers et al. (2014):** Elevated CO₂ declines in zinc, iron, and protein disproportionately impact wheat and rice, confirming their vulnerability in our nutritional weighting.
- **Ray et al. (2015):** Climate-explained yield variability checks confirm that rice exhibits the lowest climate variability share among the four crops.

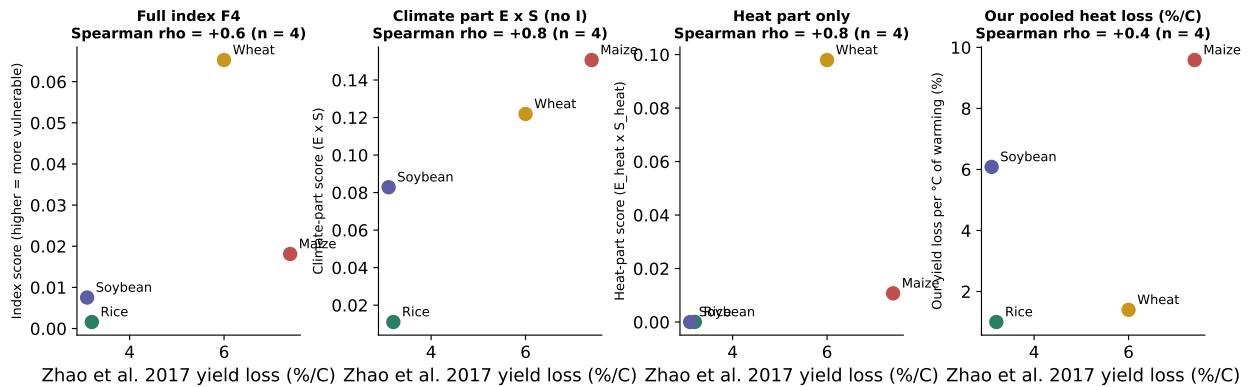


Figure 9: Crop vulnerability index scores compared with Zhao et al. (2017) global yield loss per °C.

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